

Python Exercise for lecture 5

Start-up code

```
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt

# Nice-to-have: reproducible random numbers
np.random.seed(42)
```

Key SciPy/NumPy tools you'll use

- `norm.cdf(x, loc= $\mu$ , scale= $\sigma$ )` → CDF: $P(X \leq x)$
- `norm.ppf(q, loc= $\mu$ , scale= $\sigma$ )` → quantile (inverse CDF)
- `np.random.normal(loc= $\mu$ , scale= $\sigma$ , size=n)` → simulate normal data
- Boolean masks, e.g. `mask = (x >= 95) & (x <= 105)`
- Percentages: `100 * count / total`

Uncertainty in Electrical Components: Resistor Production

In the production of $100\ \Omega$ resistors, the resistance of each resistor is modeled as a **normally distributed** random variable

- Mean (μ) = $100\ \Omega$
- Standard deviation (σ) = **unknown or given per task**

Resistors are sorted into categories:

- **5% tolerance:** within $\pm 5\%$ of $100\ \Omega \rightarrow [95\ \Omega, 105\ \Omega]$
- **10% tolerance:** between $\pm 5\%$ and $\pm 10\% \rightarrow [90\ \Omega, 95\ \Omega) \cup (105\ \Omega, 110\ \Omega]$
- **Out-of-spec:** outside $\pm 10\% \rightarrow (<90\ \Omega) \cup (>110\ \Omega)$

1) Analytical calculation (use $\sigma = 5\ \Omega$)

Goal: Using the **CDF**, compute the percentage of resistors in each category.

2) Simulation with 1000 Resistors

Goal: Simulate 1000 resistor values using `np.random.normal`.

Use **boolean masks** (True/False arrays) to check which resistors fall into each tolerance range ($\pm 5\%$, $\pm 10\%$, out-of-spec).

3) Tolerance $\rightarrow$ Standard deviation

Goal: Suppose the **5% package** should contain **half** of all resistors.

Find σ such that $P(95 \leq X \leq 105) = 0.5$ when $X \sim \mathcal{N}(100, \sigma^2)$.

4) Visualization

Goal: Using your **updated σ** from Task 3, simulate again and plot:

1. **PDF approximation:** histogram normalized to density
2. **Empirical CDF:** sort and plot cumulative proportions
(Optional) overlay the **theoretical CDF**.